

lectures on Multiphysics Modeling

Antonio De Simone

Fluids 3

24-30-31/3/26 

Consul session on 10/4 ?

Lectures 3-4 Navier-Stokes equations: Some simple solutions

o) Archimedes

1) Couette

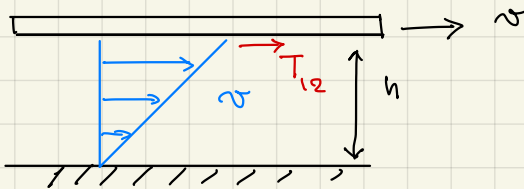
2) Poiseuille

o) Hydrostatics and the principle by Archimedes (principle of buoyancy)
~ 212 BC

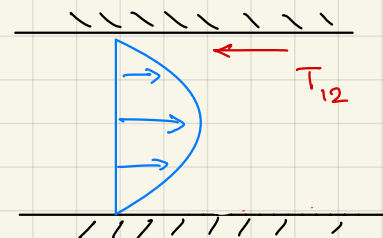
Find the hydrostatic distribution of pressure and prove Archimedes

principle: a body submerged in a fluid at rest experiences an upward force of magnitude equal to the weight of the displaced fluid.

1) Couette



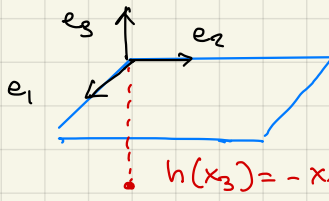
2) Poiseuille



Archimedes continued (solution)

Balance of linear momentum reduces to force balance $\rho \underline{\dot{u}} = \mu \Delta \underline{u} - \nabla p - \rho g \underline{e}_3 = 0$

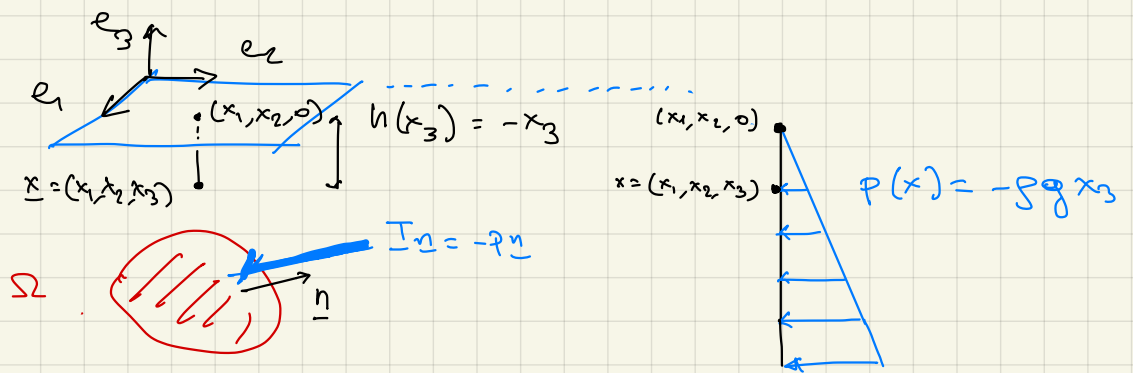
$$\nabla p = \begin{bmatrix} \frac{\partial}{\partial x_1} p \\ \frac{\partial}{\partial x_2} p \\ \frac{\partial}{\partial x_3} p \end{bmatrix} = -\rho g \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow p = p(x_3) \text{ and by integration } p(x_3) = p(0) - \rho g x_3$$



$x_3 = 0, p(0) = p_{atm} = 0 \Rightarrow p(x_3) = p(0) - \rho g x_3$

$h(x_3) = -x_3 > 0$ depth = height of water column $= \rho g h(x_3)$

Archimedes



$\underline{x} = (x_1, x_2, x_3)$ $h(x_3) = -x_3$ $\underline{T} \cdot \underline{n} = -p \underline{n}$ $p(x) = -\rho g x_3$

$$\underline{F} = \int_{\partial \Omega} \underline{T} \cdot \underline{n} \, dA \stackrel{(*)}{=} \int_{\Omega} \operatorname{div} \underline{T} \, dV = \int_{\Omega} (-\nabla p) \, dV = \int_{\Omega} \rho g \underline{e}_3 \, dV$$

$$= \rho g \operatorname{vol}(\Omega) \underline{e}_3$$

where we have used $\underline{T} = -p \underline{I}$ so that $\operatorname{div} \underline{T} = \operatorname{div} (-p \underline{I}) = -\nabla p$

$\operatorname{div} (-p \underline{I}) = -\nabla p$, see below

Archimedes buoyancy principle

where we have used $\underline{T} = -p \underline{I}$ so that

$$\operatorname{div} \underline{T} = \operatorname{div} (-p \underline{I}) = -\nabla p$$

$$(\operatorname{div} \underline{T})_i = \sum_j \frac{\partial}{\partial x_j} T_{ij} = \sum_j \frac{\partial}{\partial x_j} (-p \delta_{ij}) = \frac{\partial}{\partial x_i} (-p) = -(\nabla p)_i$$

(*) Alternatively: $-\int_{\partial \Omega} p \underline{n} \cdot d\mathbf{A} = -\int_{\Omega} \nabla p \cdot d\mathbf{V}$ by div. thm's.

Towards problems 1 and 2.

They both involve flows along a straight line (1D flows): many components of $\underline{v}$ vanish if we assume $\underline{e}_1$ is the flow direction

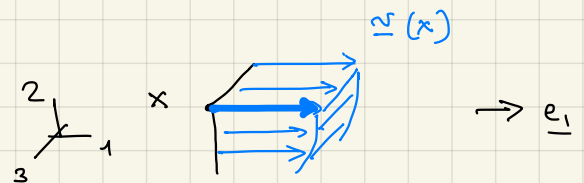
Some simple solutions for plane steady flow (1D flows), in the absence of body forces.

For steady flow in the absence of body forces (NSE) reduce to

$$\left[\begin{array}{c} \nabla_x \\ \Delta_x \\ \nabla_x \end{array} \right] \rho (\nabla \underline{v}) \underline{v} = \mu \Delta \underline{v} - \nabla p \quad ; \quad \text{div}_x \underline{v} = \nabla_x \cdot \underline{v} = 0 \quad \text{omit subscript } x \text{ for simplicity} \quad (1)$$

Consider the velocity field

$$\underline{v}(x) = v_1(x_1, x_2) \underline{e}_1$$



Then (1)₂ implies that $\frac{\partial v_1}{\partial x_1} = 0$, hence $v_1 = v_1(x_2)$ and $\underline{v}$ is simple shear.

The matrix of $\nabla \underline{v}$ is

$$\left[\nabla \underline{v} \right] = \left[\frac{\partial v_i}{\partial x_j} \right] = \begin{bmatrix} 0 & \frac{\partial v_1}{\partial x_2} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad (2)$$

$$\text{and } (\nabla \underline{v}) \underline{v} = 0.$$

$$\text{(Check: } \begin{bmatrix} 0 & \partial v_1 / \partial x_2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix})$$

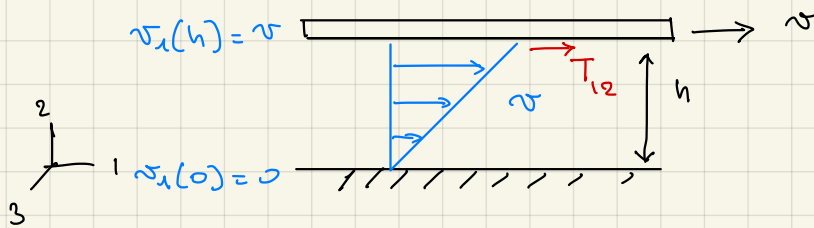
Therefore (1)₁ reduces to

$$\mu \frac{\partial^2 v_1}{\partial x_2^2} = \frac{\partial p}{\partial x_1}, \quad \frac{\partial p}{\partial x_2} = \frac{\partial p}{\partial x_3} = 0 \quad (3)$$

so that $p = p(x_1)$.

$$\implies \mu \frac{d^2 v_1}{dx_2^2} = \frac{dp}{dx_1}$$

Problem 1 : Flow between two parallel plates (Couette)

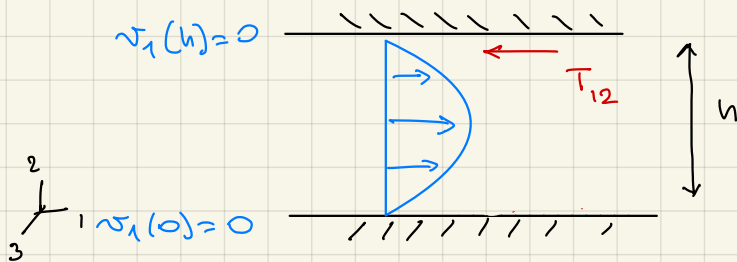


pressure $P \equiv \text{const} = P_0$

$$v_1 = v \frac{x_2}{h}$$

$$T_{11} = T_{22} = T_{33} = -P_0, \quad T_{12} = T_{21} = \mu \frac{v}{h}$$

Problem 2 Flow between two fixed parallel plates under a pressure gradient (Poiseuille)



pressure p : $\frac{\partial p}{\partial x_1} = -\delta$ constant
(pressure drop per unit length)

$$v_1 = \frac{\delta}{2\mu} x_2 (h - x_2)$$

$$T_{11} = T_{22} = T_{33} = -P; \quad T_{12} = T_{21} = \delta \left(\frac{h}{2} - x_2 \right)$$

Volume discharge $Q = \frac{\delta}{12\mu} h^3$
per unit time through
a cross-section of unit depth

Remark : Problem 1 and Problem 2 provide two independent ways to "measure" viscosity μ .

Parabolic flow of Problem 2 arises also in axisymmetric conditions (flow in a circular pipe) and is called Poiseuille flow.

Remark Problem 1 and Problem 2 are solutions also for Stokes flows, see later.

Re: $\mu \frac{d^2 v_1}{dx_2^2} = \frac{dp}{dx_1} \quad (*)$

$$T = \mu \frac{dv_1}{dx_2} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - p \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (**)$$

Problem 1

Look for a solution with pressure $p \equiv \text{const} = p_0$. From (*), v_1 is linear in x_2

and using BC's $v_1 = \frac{v}{h} x_2$

(**) becomes $T = \frac{\mu v}{h} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} - p_0 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \begin{aligned} T_{11} &= T_{22} = T_{33} = -p_0 \\ T_{12} &= T_{21} = \mu \frac{v}{h} \end{aligned}$

Problem 2

Constant pressure leads to trivial $v \equiv 0$ because of Problem 1 and BC's

Non constant p : (*) $\Rightarrow \mu \frac{d^2 v_1}{dx_2^2} (x_2) = \frac{dp}{dx_1} (x_1)$ so $\frac{dp}{dx_1}$ is constant.

Set pressure p s.t. $\frac{\partial p}{\partial x_1} = -\delta$ constant pressure drop per unit length

From (*) $\frac{d^2 v_1}{dx_2^2}$ is constant, so that v_1 is quadratic in x_2 , with BC's

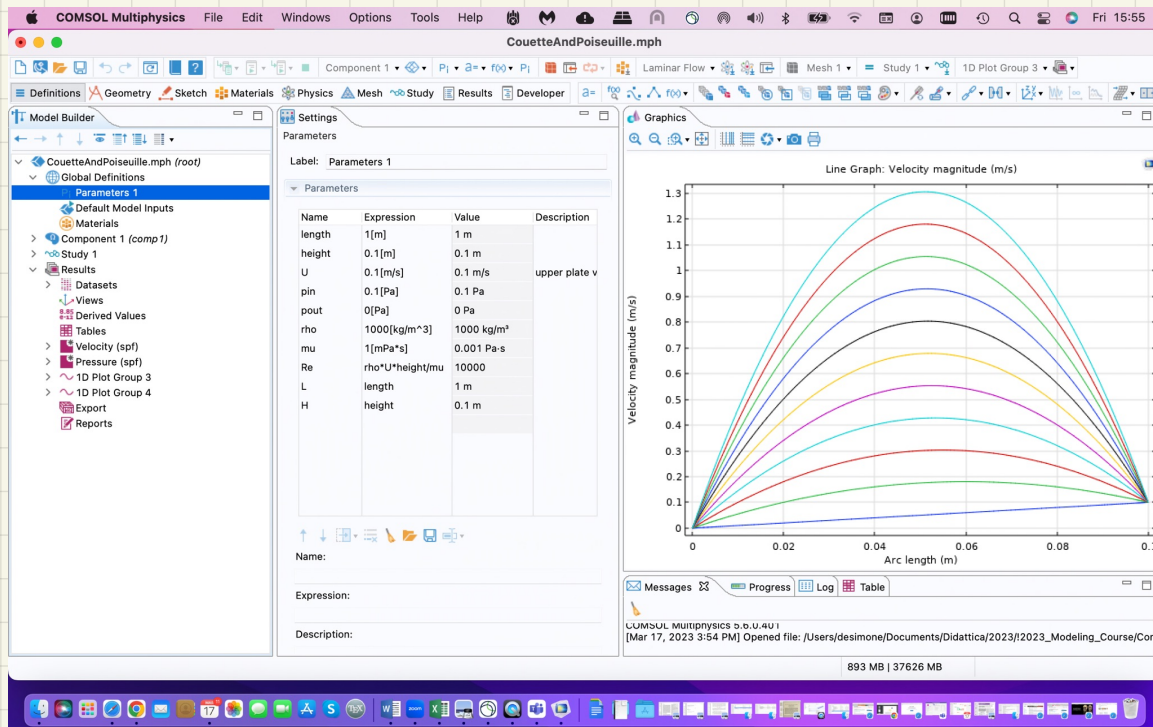
$$v_1 = \frac{\delta}{2\mu} x_2 (h - x_2)$$

$$\frac{dv_1}{dx_2} = \frac{\delta}{2\mu} (h - x_2 - x_2) = \frac{\delta}{\mu} \left(\frac{h}{2} - x_2 \right)$$

From (**) $T_{11} = T_{22} = T_{33} = -p$, $T_{12} = T_{21} = \delta \left(\frac{h}{2} - x_2 \right)$

$$Q = \int_A v \cdot n \, dA = 1 \int_0^h v_1(x_2) dx_2 = \dots = \frac{\delta}{12\mu} h^3$$

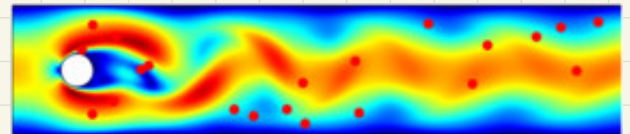
Now move to COMSOL files.



1. Couette and Poiseuille

2. Poiseuille

3. cylinder flow (von Korman vortex sheet)



Case 3 is particularly relevant. (shedding of vortices from a fixed obstacle)

It shows that NSF can have time-dependent solutions under time-independent data (forcing terms), besides the "trivial" time-independent one. This is an instability, made possible by the fact that (NSF) are nonlinear, due to convective acceleration $(\nabla \cdot \mathbf{v}) \mathbf{v}$.

